

YINGAO (ELAINE) YAO

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<https://elaineyao.github.io/>

RESEARCH INTERESTS

My research interests are at the intersection of Software Engineering and Machine Learning (ML). I am particularly interested in developing software testing techniques (e.g., fuzzing) to improve the security, reliability, and correctness of ML systems.

EDUCATION

Cornell University Ph.D. in Computer Science. Advisor: Dr. Saikat Dutta.	<i>September 2024 - Present</i>
University of British Columbia MSc. Electrical and Computer Engineering. Advisor: Dr. Karthik Pattabiraman.	<i>September 2021 - December 2023</i>
University of Electronic Science and Technology of China B.E. Electrical and Communication Engineering GPA: 3.96/4.0	<i>September 2017 - June 2021</i>

HONORS AND AWARDS

IEEE DSN Student Travel Grant Award	<i>2023</i>
Faculty of Applied Science Graduate Award, University of British Columbia	<i>2022</i>
Outstanding Graduate, Sichuan Province	<i>2021</i>
Outstanding Winner, COMAP Interdisciplinary Contest in Modeling	<i>2020</i>
Thanksgiving Scholarship for Modern Scientists	<i>2020</i>
National Scholarship	<i>2020</i>
National Second Prize in China Undergraduate Mathematical Contest in Modeling	<i>2018</i>

PUBLICATIONS

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman. IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).	<i>DSN 2023</i>
Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms. Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman. ACM SIGSAC Conference on Computer and Communications Security (CCS).	<i>CCS 2023</i>
Master thesis: Framework for discovering GPS spoofing attacks in drone swarms Drone Swarms. Yingao (Elaine) Yao. University of British Columbia.	<i>UBC 2025</i>

PREPRINTS

A Regression Testing Framework with Automated Assertion Generation for Machine Learning Notebooks (Under submission)	<i>arXiv 2025</i>
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Yingao (Elaine) Yao, Vedant Nimje, Varun Viswanath and Saikat Dutta.
<https://arxiv.org/abs/2509.13656>.

ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic (Preprint) *arXiv 2023*
Gargi Mitra, Pritam Dash, **Yingao Elaine Yao**, Aastha Mehta, and Karthik Pattabiraman.
<https://arxiv.org/abs/2312.06140>.

WORKING EXPERIENCE

Security Researcher
Fortinet, Vancouver, Canada.

Jan 2024 - Aug 2024

TEACHING EXPERIENCE

Teaching Assistant for CS 5150: Software Engineering
Cornell University, Ithaca, NY, USA

Spring 2025

Teaching Assistant for CPEN 321: Software Engineering *Spring 2022, Fall 2022, Spring 2023*
University of British Columbia, Vancouver, Canada.

ACADEMIC SERVICE

Artifact Evaluation Program Committee

2024

19th USENIX Symposium on Operating Systems Design and Implementation (OSDI).

European Conference on Object-Oriented Programming (ECOOP).

18th USENIX WOOT Conference on Offensive Technologies (WOOT).

ACM Conference on Computer and Communications Security (CCS).

USENIX Annual Technical Conference (USENIX ATC).

The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Artifact Evaluation Program Committee

2023

The 34th IEEE International Symposium on Software Reliability Engineering (ISSRE).

DEPARTMENT SERVICE

Department of Computer Science, Picnic Czar
Cornell University, Ithaca, NY.

Fall 2025

MENTORING

Vedant Ramesh Nimje (BS, Veermata Jijabai Technological Institute, Mumbai)

Varun Viswanath (BS, Dwarkadas J Sanghvi College of Engineering)

Parshwa Nandu (BS, Dwarkadas J Sanghvi College of Engineering)

PRESENTATIONS

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms

DSN 2023

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms.

CCS 2023

OPEN-SOURCE CONTRIBUTIONS

SwarmFuzz: <https://github.com/DependableSystemsLab/SwarmFuzz>

NBTest: <https://github.com/seal-research/NBTest>

SKILLS

Tools

AFL, LibFuzz, Python, C/C++, Bash

Spoken Languages

English (Fluent), Mandarin (Fluent)